

WORKSHEET 2: THE RESEARCH PROBLEM

1. THE VERTEX SET OF AN EDGE

Previously, we talked about $V(G)$ to be the set of all of the vertices of graph G . It will also be helpful for us to define $V(a \text{ subset of edges})$.

Consider the graph



Note. This is a special type of graph called a *linear graph* (sometimes *path graph*). In particular, this is L_6 .

Then, we can define $V(\{BC, CD\}) = \{B, C, D\}$. This is the set of vertices that touch the edges BC and CD . When we look at the vertex set for just one edge, like $V(\{BC\}) = \{B, C\}$, we drop the brackets for convenience, i.e., we can write $V(BC) = \{B, C\}$.

2. EDGE BLANKETS

Let G be a graph with vertex set $V(G)$ and edge set $E(G)$.

DEFINITION. A subset of edges $J \subseteq E(G)$ is an **edge blanket** if for all $i \in E(G)$, we have $V(i) \cap V(J) \neq \emptyset$.

Meaning, J is an **edge blanket** if for every edge in our graph G , an edge in J touches it.

Consider the graph



Then, $J = \{AB, CD, EF\}$ is an edge blanket, because AB touches the edge BC and CD touches the edge DE .

Exercise 1. Is $J = \{BC, CD, DE\}$ an edge blanket?

Exercise 2. Is $J = \{AB, BC, CD\}$ an edge blanket?

Exercise 3. Is $J = \{AB, BC, CD, DE, EF\}$ an edge blanket?

Exercise 4. For any graph G , is $E(G)$ an edge blanket?

3. INTERNAL EDGES

Let G be a graph with vertex set $V(G)$ and edge set $E(G)$.

DEFINITION. A subset of edges $J \subseteq E(G)$ has an **internal edge** if there exists an $i \in J$ that is internal, i.e., $V(i) \subseteq V(J \setminus \{i\})$.

Example. In the above example, $J = \{AB, BC, CD\}$ has an internal edge, BC .

We see $J \setminus \{BC\} = \{AB, CD\}$. So $V(J \setminus \{BC\}) = \{A, B, C, D\}$. Further, $V(BC) = \{B, C\}$. So we see $\{B, C\} \subseteq \{A, B, C, D\}$.

Exercise 5. Does $J = \{AB, CD, EF\}$ have an internal edge?

4. d -EDGE DENSE

DEFINITION. Let d be a positive integer. A graph G is **d -edge dense**, if for all subsets $J \subseteq E(G)$ of size d , J is an edge blanket or it has an internal edge.

We will now determine whether G is 3-edge dense, i.e., when $d = 3$.

Exercise 6. Write down all of the set of edges of G of size 3.

Exercise 7. For each set, determine whether it is an edge blanket or has an internal edge.

Exercise 8. Is G 3-edge dense?

We want to understand what graphs are d -edge dense. What are some ways we can systematically look at graphs? What are the easiest graphs we can look at first?

Exercise 9. For any graph G , is G $|G|$ -edge dense? Justify.

5. YOUR GOAL

Your goal for this summer is to find as many d -edge dense graphs as you can (for different d). You have some freedom, but we would recommend starting with the graphs with fewer vertices first.

We also recommend you think about what kinds of graphs have the possibility of being 1-edge dense and which ones have the possibility of being $|V(G)|$ -edge dense.

Once you've generated several examples of d -edge dense graphs, you should begin to formulate hypotheses that you think a graph might have to satisfy for a graph to satisfy the d -edge dense property.